

HOW CAN THE UNIVERSE BE INFINITE?

It is difficult to imagine something going on forever. That is why ancient astronomers believed that, somewhere far beyond the stars, there was an end to the universe. They thought there must be an edge marking where the universe ceased to be. Now we know that, however far and fast you travel through the universe, you would never reach an end. Some theories of the universe suggest that it has only a limited volume, which means that you might return to your starting point if you traveled long enough.

WHY

CAN'T THERE BE AN EDGE TO THE UNIVERSE?

The ideas of a finite and infinite universe both posed problems for the ancient Greeks. The concept of something infinite was impossible to understand, and yet, if the universe ended at some point, what, if anything, would you see if you looked over the edge? Anything on the other side must, by definition, belong to the universe, as the universe is all there is. The expanding universe, a new idea based on modern physics, solved these problems that had troubled scientists right up until the early 1900s.

